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Borane ($B_m H_n$), Hydrogen rich, Proton Boron fusion fuel materials for high yield laser-driven Alpha sources

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Borane (B_mH_n), Hydrogen rich, Proton Boron fusion fuel materials for high yield laser-driven Alpha sources

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ABSTRACT: We propose for the first time, a new fuel-material for laser-driven Proton Boron (P-B) fusion nuclear reactions. We propose, Hydrogen rich, Borane (B_mH_n) materials as fusion fuel as compared to Boron Nitride (BN) presently used. We estimate more than 10-fold increase in the yield of nuclear fusion reactions, and Alpha-particle flux, when, for example Ammonia Borane (BNH_6) laser-target material will be used compared to the state of the art normalized flux $\sim 10^8$

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Alphas/sr/J from BN targets. BNH_6 contains $\sim 1000\times$ higher concentration of Hydrogen compared to BN. We report the manufacture of the first solid-pellets Ammonia Borane laser-targets. To obtain high Flux Alpha sources from repetitive lasers we propose new BNH_6 target geometries: liquid (molten) droplets/jets; or translated tape- or disc-targets coated with BNH_6 powder. Targets would be irradiated in low pressure, ambient buffer gas. To enhance the fusion/Alpha yield of ultra-high intensity PetaWatt laser-target interaction we propose nano- and micro-structured Borane targets. As applications, we propose to use the Alpha-driven nuclear reactions inside the laser-driven Borane targets for new schemes to produce short-lived medical radioisotopes. Such laser-driven radioisotope beamlines would be installed directly in hospitals. Borane materials, like Diborane (6), B_2H_6 , are also proposed as nuclear-fuels for laser-driven Proton-Boron fusion energy generation. The high dilution of Boron in Hydrogen $B/H = 33\%$ would need to be further enhanced to $B/H < 15\%$ to cut radiation losses from the hot and dense fusion pellet.

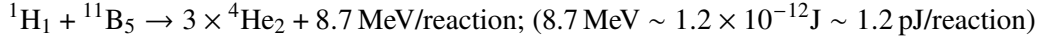
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1 Introduction

The Proton-Boron (P-B) fusion reaction is also referred to as HB11 fusion since it fuses a proton with a Boron nucleus in a strong exothermal reaction (8.7 MeV/reaction) which emits three charged particles: energetic alpha-particles (Helium nuclei, ${}^4\text{He}_2$) and no neutrons:



The maximum crosssection for the reaction is obtained when the colliding particles have a kinetic energy of $\sim 600 \text{ keV}$ which is more than $10\times$ larger than the $\sim 50 \text{ keV}$ for Deuterium Tritium (DT) fusion.

Surprisingly, scientists have shown, in the last two decades that such fusion reactions can be driven by readily available lasers. Figure 1. shows the rapid increase of Alpha-particle flux from such laser-driven boron targets, for example in [1–3] which use low repetition rate, high energy laser pulses. A first experiment to demonstrate average flux of Alpha particles used a high repetition rate, kHz table-top laser to generate kHz Alpha-particles emission [4].

In the future, compact laser-driven Alpha-sources could improve the logistics of medical radioisotopes production and delivery to the patient, for medical imaging and therapy [5, 6]. Longer [days] lived isotopes could be produced in a central facility and quickly distributed to hospitals. Shorter lived [hundreds of minutes or less] isotopes could be manufactured inside a laser-driven Alpha Source hospital facility and delivered promptly to the patient. Another key application, Nuclear Fusion Energy generation [7–13] from Proton-Boron Fusion reactions would be extremely beneficial because: the P-B reaction fusion energy is emitted as kinetic energy of charged (Alpha) particles which could be converted to electricity by applying Electric or Magnetic field; the P-B fusion reaction produces very low radioactivity due to absence of primary neutron generation (aneutronic reaction).

Laser-Produced-Plasma induced Proton-Boron-Fusion alpha-particle emission in excess of 10^{10} alphas/shot have been demonstrated experimentally from ‘in-target’ laser irradiation of solid Boron targets containing traces of Hydrogen, for example in [1–3] and references within. The state of the art in laser-driven Alpha-sources generated from Proton-Boron fusion reaction is the record Alpha-flux with $>10^{10}$ alphas/sr/pulse [1] obtained with targets provided by one of our colleagues, AP, from FBK, Italy: solid Boron Nitride (BN). BN contains traces of Hydrogen $\sim 1\%$ which is incorporated during its manufacture. Boron Nitride is obtained by reacting trioxide (B_2O_3) or boric acid (H_3BO_3) with ammonia (NH_3) or urea ($\text{CO}(\text{NH}_2)_2$) in a nitrogen atmosphere [1, 2]. FBK also provides hydrogen-doped boron targets for P-B fusion Alpha sources.

Figure 1 summarizes the state-of-the-art experiments achieving high Alpha-particle-fluxes from laser-driven Proton-Boron reactions:

- (a) Using LFEX Laser: E-laser = 1400 J, T-laser = 2.2 ps, I-laser $\sim 3 \times 10^{19} \text{ W/cm}^2$, the obtained Alpha flux is $\sim 3 \times 10^{10}$ Alphas/sr/pulse and the Alpha normalized flux $\sim 10^7$ Alphas/sr/J [2, 3].
- (b) Using PALS Laser diver: E-laser = 600 J, T-laser = 300 ps, I-laser $\sim 3 \times 10^{16} \text{ W/cm}^2$, the obtained Alpha Flux is $\sim 3 \times 10^{10}$ Alphas/sr/pulse and Alpha normalized flux $\sim 5 \times 10^7$ Alphas/sr/J [1].

Record high repetition rate alpha flux of 10^6 alphas/sec using a ‘table-top’ laser-driver with low pulse energy $E = 20 \text{ mJ/pulse}$, high repetition rate $f = 1 \text{ kHz}$ pulses focused on target to an intensity of $\sim 2.3 \times 10^6 \text{ W/cm}^2$ [4]. The target consisted of thin film of poly(ethylene) deposited on

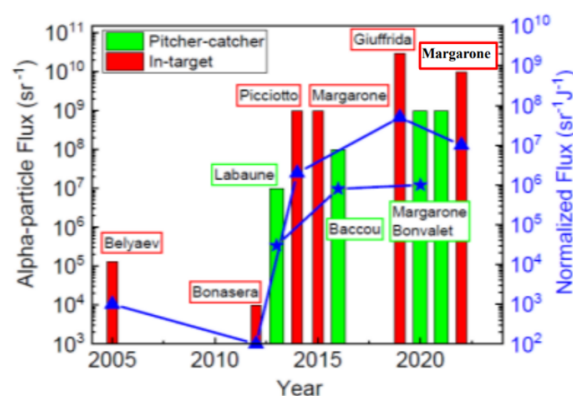


Figure 1. The rapid increase in the Alpha-particle-flux obtained from laser-driven Proton-Boron reactions over the last ten years. For the ‘in-target’ interaction the laser is focused directly on the Boron-containing target. For ‘pitcher-catcher’ interaction the laser is focused on a plastic foil-target placed in front of the Boron target. The laser accelerated protons at the back of the plastic foil collide with the Boron nuclei in the target and induce Proton-Boron fusion reactions. Blue triangles/stars refer to the right axis for Normalized Flux. Reproduced from ref. [2]. CC BY 4.0. D. Margarone et al., *In-Target Proton-Boron Nuclear Fusion Using a PW-Class Laser*, *Appl. Sci.* **12** (2022) 1444, figure 1 modified.

Boron Nitride (BN) substrate which was translated between laser pulses such that a fresh target is irradiated by the next laser pulse.

We proposed in October 2022, for the first time, at the P-B Fusion Workshop [14], a new class of Hydrogen rich, chemical-molecules, Boranes (BnHm) [15] to be used as P-B fusion laser-target materials as well as new designs for such targets — this paper is based on our presentation. We expect the Hydrogen rich target materials to further enhance the number of laser-driven Proton-Boron reactions, compared to presently used Boron Nitride (BN) targets, and therefore the laser energy conversion to Alpha-particle emission by more than 10-fold (section 2). This would enhance the present state of the art normalized Alpha flux from $\sim 10^8$ Alphas/sr/J to $> 10^9$ Alphas/sr/J. In section 3 we present general properties of Boranes relevant to Alpha Source generation and to nuclear fusion fuels in general. For example Ammonia Borane (BNH₆) has a Hydrogen/Boron ratio of $H/B = 600\%$ which is 1000× higher than in Boron Nitride targets with $H/B < 1\%$. In section 4 we report for the first time manufactured Ammonia Borane solid-pellet targets obtained by compressing the powder of Borane micro- and nano-crystals into a pellet. The surface of such pellet targets appears to be randomly micro-structured.

In section 5 we propose further increasing the average flux of Alpha particles by new, proposed high repetition rate Borane targets for high repetition rate lasers. Proposed are liquid (molten) droplet- or jet-targets as well as translating tape- or disc-targets. Target debris reduction techniques are discussed, like thin targets and laser-target irradiation in the presence of buffer gas. In section 6, we propose new Borane nano-structure targets for femtosecond PW laser irradiation and micro-structure targets for PW picosecond laser irradiation. Such targets will absorb efficiently the ultra-high intensity ultrahigh power Petawatt lasers. Such ultra-high laser intensities and fields would accelerate the target particles to large energies and further enhance the laser-to-alpha-particle conversion efficiency.

In section 7.1 we propose new schemes for medical radioisotope production within laser-driven Borane targets, using the in-target laser-induced nuclear reactions. In section 7.2 we propose to use Borane materials as nuclear-fuel for Proton-Boron fusion energy production schemes.

2 Estimate of more than 10-fold enhanced yield of Proton-Boron fusion reactions and Alpha-particle emission from Ammonia Borane compared to Boron Nitride laser-target-materials

We expect the number of Proton-Boron (P-B) Fusion reactions in the laser-irradiated target to scale with the proton density in the Boron target. If we choose Ammonia-Borane (BNH₆) material for example, the Hydrogen to Boron ratio is $H/B = 600\%$ compared to the state of the art BN targets with $H/B < 1\%$.

Let us estimate the dependence of the number of P-B reactions on the proton (Hydrogen ion) density in the target. If a beam of N_p protons impinges on a Boron target with density n_B the total number of Proton-Boron fusion reactions is:

$$N_{PBF} = N_p \times \text{Sigma (cm}^2\text{)} \times \text{Lambda (cm)} \times n_B \text{ (cm}^{-3}\text{)}$$

where: $[N_{PBF}]$ is the number of P-B Fusion Reactions; $[N_p]$ is the number of protons in beam; $[\text{Sigma}]$ is the P-B fusion crosssection, in cm^2 ; $[\text{Lambda}]$ is the protons range in boron, in cm; $[n_B]$ is the Boron atoms number density in target, in cm^{-3} .

This simple estimate points out that in this approximation the number of P-B reactions is proportional to the number of protons in the beam. Therefore the larger the proton number the larger number of P-B fusion reactions hence larger number of Alpha-particles emitted.

A similar approximate dependence would hold for ‘In-Target’ P-B reactions where the protons are accelerated by the in-target laser-generated fields. Here, the number of ‘in-target’ accelerated protons (N_p) would depend on the Hydrogen number density and an ‘acceleration volume’ in which the protons are accelerated by the in-target laser fields.

We therefore expect the laser to Alpha-flux conversion efficiency to increase proportional to the density of Hydrogen atoms in the target, in the case of ‘in-target’ laser-driven fusion reactions. This is the main motivation of proposing the Hydrogen-rich Borane materials for targets for laser-driven P-B fusion Alpha sources.

If we choose the Hydrogen-rich Ammonia-Borane (BNH₆) material for example, the Hydrogen to Borane ratio is $H/B = 600\%$ compared to the state of the art BN targets with a ratio of $H/B < 1\%$. This $\sim 1000\times$ higher Hydrogen density would imply a dramatically enhanced fusion yield of ~ 1000 -fold for Ammonia Borane target material. Below we consider the assumptions made in our simple calculation, and propose an estimate of only more than 10-fold enhancement in fusion yield.

In our simple calculation we assumed 3-Dimension interaction of accelerated ions inside the target. The acceleration mechanism inside laser-targets is not well understood at the moment. There is evidence of laser penetration inside special targets, like nano- or micro-structured Borane targets. We describe such targets in section 6 and section 4.

If the ions are accelerated only in 2D, in the plasma generated in the laser focus, we would only expect a more than 10-fold increase in the yield of P-B fusion reactions compared to BN targets. Throughout this paper we take the conservative expectation of more than 10-fold increase in fusion the yield in BNH₆ laser-targets compared to BN targets.

We plan to carry out future experiments to investigate the scaling of the Alpha particles yield as a function of Hydrogen concentration in the Boron targets.

3 New, proposed Borane target materials for laser driven Proton Fusion reaction and Alpha generation

In our view, the most attractive material which is rich in both Boron and Hydrogen and having a large ratio of Hydrogen/Boron (H/B) is the class of molecules called Boranes (BnHm) [15] or Boron Hydrides. There are a large number of molecules in the Borane family. Boranes are in gas or solid form. For example Diborane (6) with chemical formula (B_2H_6) has the ratio $H/B = 300\%$. It is a gas in normal conditions, becomes a liquid at $\sim -92.5^\circ\text{C}$ and solid at $\sim -164.9^\circ\text{C}$. Borane molecules with increasing number of Boron and Hydrogen atoms are for example: tetraborane (10) with formula B_4H_{10} , also a gas; Decaborane (14): $B_{10}H_{14}$, solid with density of 0.94 g/cm^3 and melting temperature of $\sim 98^\circ\text{C}$; or $B_{18}H_{22}$. Boranes are gases or solids in normal conditions. The solids appear in the form of micro-and-nano-crystalline powders, have densities around $\sim 1\text{ g/cm}^3$ and low melting points of around $\sim 100^\circ\text{C}$. Most boranes are soluble in common solvents. For example Decaborane (14) is soluble in non-polar solvents and even slightly soluble in cold water.

Examples of other interesting Borane compounds for Proton-Boron fusion laser-target materials are: Ammonia-Borane (also called Borazane), BNH_6 (or H_3BNH_3 in another common notation) or Lithium-dodeca-closo-dodecaborate $Li_{12}B_{12}H_{12}$ salt. Both have appearances of micro-crystalline powders and low melting points. Most Boranes are toxic and flammable although some are quite benign, like Ammonia Borane which we have chosen as our main target material.

Our proposed, main target material, Ammonia Borane (BNH_6), CAS Number 13774-81-7, appears as colourless crystals, has a density of 0.78 g/cm^3 , and a very low melting point of 104°C . The Ammonia Borane powder appears to be safe to handle according to the instructions from the suppliers: [Sigma Aldrich]: No toxicity and No flashpoint, dust protection needed; [American Elements]: NONH for all transport.

Figure 2 shows a picture of the Ammonia Borane crystals powder as supplied by one of the vendors: Stanford Advanced Materials. The powder contains micro- and nano-crystals. We propose to use such crystalline powder as a raw material for our P-B fusion laser-targets. It is interesting that some vendors even offer Ammonia Borane powder containing only nano-crystals. We consider such powder a potential cost-effective material for nano-structured laser-targets (sections 4, 5 and 6). Such micro- and nano-crystal powders could be investigated as ‘random’ micro- and nano-structures laser targets (section 6). We also propose for such microscopic, cylindrical mono-crystals to be separated from the powder and made into stand-alone laser targets. More details and pictures on Ammonia Borane powder can be seen on the vendor’s website: www.samaterials.com/boron/2723-ammonia-borane.html. Catalogue BC2723, Molecular formula BH_6N , CAS Number 13774-81-7, Appearance: Beige crystalline solid, Purity 99%–99.999%.

4 New, manufactured, Ammonia Borane pellet-targets for laser-driven Alpha-sources from Proton-Boron fusion

We now report, for the first time, the successful manufacturing of Ammonia Borane pellets for laser targets for Proton-Boron fusion. One of the authors (ICET) suggested compressing the Ammonia Borane powder in order to obtain solid pellet laser-targets. Our co-author (CS) manufactured, for the first time, the Ammonia Borane pellets shown in figure 3(a), by compressing the nano-micro-crystalline

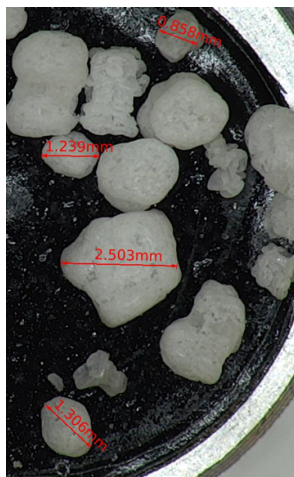


Figure 2. Ammonia Borane crystals powder from the Stanford Advanced Materials. Microscope picture taken by our co-author CS and his team. Note cylindrical microcrystals, approximately 100 micrometres diameter and smaller, forming the powder boulders. Such microscopic, cylindrical mono-crystals could be separated from the powder and made into stand-alone laser targets. More details and pictures on Ammonia Borane powder can be seen on the vendor’s website: www.samaterials.com/boron/2723-ammonia-borane.html. Catalogue BC2723, Molecular formula BH_6N , CAS Number 13774-81-7, Appearance: Beige crystalline solid, Purity 99%–99.999%.

powder material purchased from Stanford Advanced Materials shown in figure 2 above. The pellet has standard dimensions: Diameter = 12 mm and Thickness = 1.2 mm.

A $5\times$ magnified optical microscope image of the surface (figure 3(b)) shows a rough, microstructured surface.

Figure 3(c) shows the pellet surface scanned with profilometer with a $5\text{ }\mu\text{m}$ tip radius. The surface structure line out has an average roughness of around $200\text{ nm} = \text{Ra}$ and the largest peak to valley is around $1.5\text{ }\mu\text{m}$.

Figure 3(d) shows images of the Borane pellet-target obtained with a white-light optical. The images show the surface roughness of the ammonia borane pellet. They show some nice microstructure of the order of a few microns — this is overlaid on a more surface waviness that is probably down to the press (used in manufacturing) and is larger scale — about $5\text{--}10\text{ }\mu\text{m}$ over the area of a few mm.

Images in figure 3 indicate that the pellet surface appears to have structure on the micrometre scale. This can be explained by the fact that the Borane powder contains Borane micro- and nano-crystals (figure 2). The target surfaces also macroscopic have imperfections. We will keep improving the technique to obtain more uniform, surfaces, while still maintaining the micro-structure. Even the existing surfaces can be irradiated with laser if we choose the appropriate surface area.

A micro-structured pellet surface would be beneficial in maximizing the absorption of the laser-driver in the target surface, especially if ultrahigh laser intensities are required (section 6). The micro-structured surface would also be beneficial for deeper penetration of the laser Electro-Magnetic field which could lead to higher fusion yield (section 2). It looks like nature may appear to offer us this very simple technique to obtain, a micro-structured, Ammonia Borane target surface, by compressing nano- and micro-crystalline commercial powder.

In the next experiments we aim to use such Hydrogen-rich ($\text{H/B} = 600\%$) Ammonia Borane solid pellets as targets for laser-driven Proton-Boron fusion alpha generation.

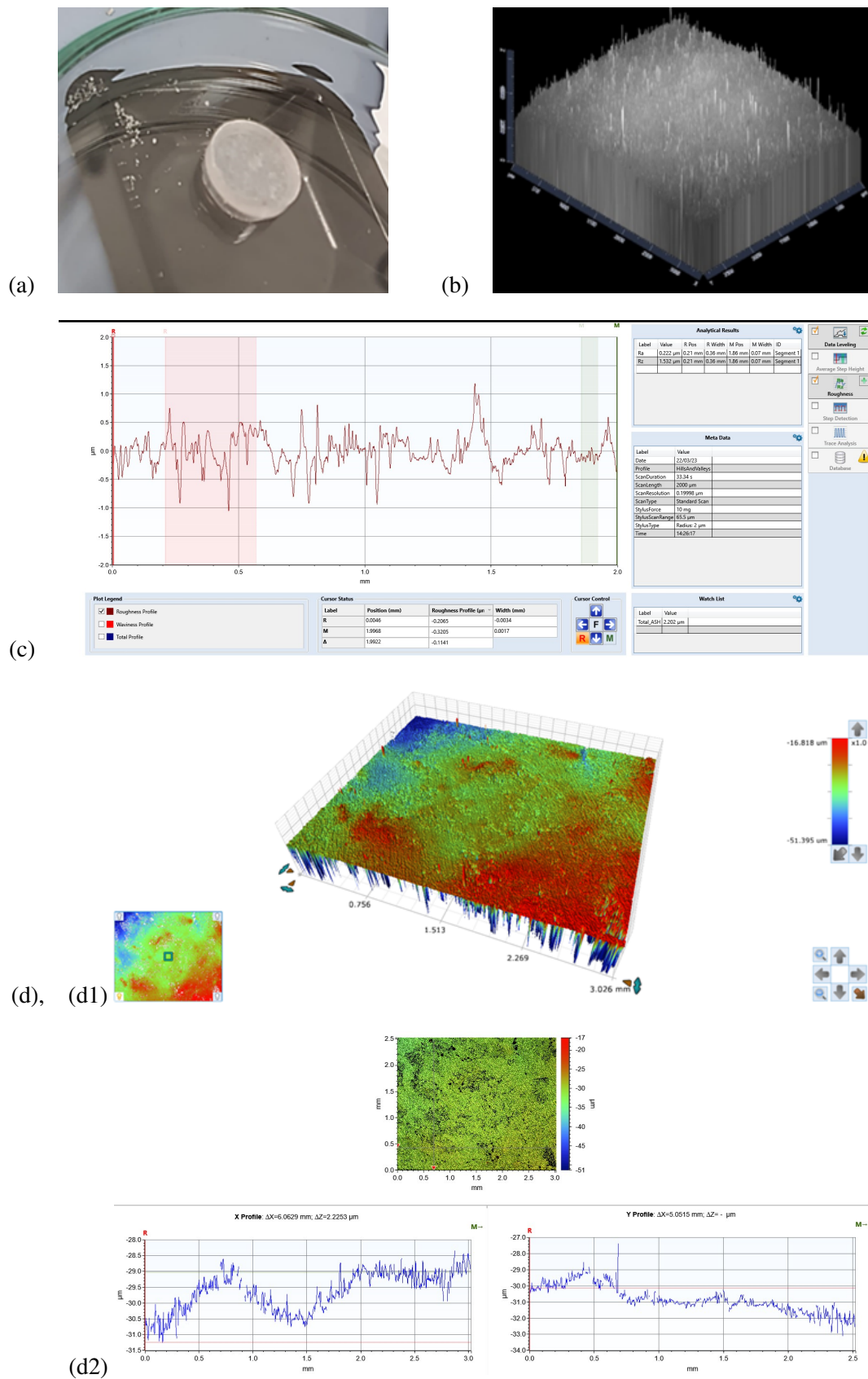


Figure 3. Continued on the next page.

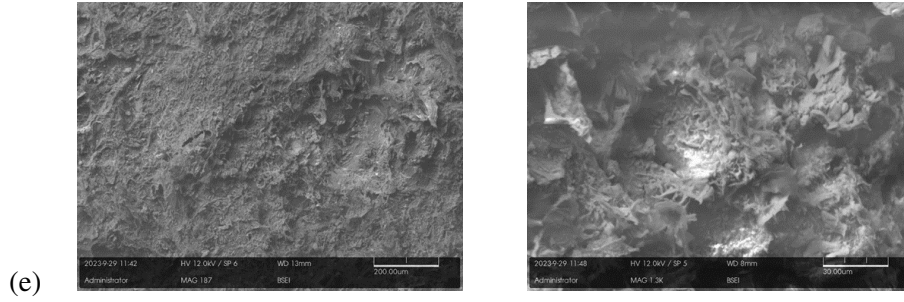


Figure 3. (a) Ammonia-Borane pellet as laser-target for Proton-Boron Fusion. Diameter = 12 mm, Thickness 1.2 mm — manufactured for the first time by one of our authors, CAS, by compressing the powder material as proposed by one of our authors, ICET. Images (a), (c) (d) and (e) obtained by our co-author, CAS; (b) Microscope image of the Pellet surface. Scales are in micrometres. The image was taken on Zeiss optical microscope, model: Axio Imager.A2m. with $\times 5$ objective. Image obtained by Dr Katarzyna Batani at the CLPU facility; (c) The pellet surface was scanned with a surface profilometer (Bruker Dektak) with a $5\mu\text{m}$ tip radius. The surface structure line out has an average roughness of around $200\text{ nm} = R_a$ and the largest peak to valley is around $1.5\mu\text{m}$. (d) optical interferometer (Bruker Contour White Light interferometer) image of the surface of pellet surface: (d1) 3-d map of a central region of the pellet, as shown on the left image. The 3D image has a tilt from bottom right to top left but that is an artefact of the scan not being levelled. (d2) scan of the interferometer image at ‘site-2’ as indicated on the image. (e) Backscatter Electron Image (BSEI) of the surface of the BNH_6 pellet-target.

5 Proposed, new Ammonia Borane molten-droplet and tape targets for high repetition lasers driven Alpha-particle sources from P-B fusion

High fluxes of Alpha-particle radiation, e.g. 10^{11} Alphas/second and above, are of great interest in applications like the production of radioisotopes for medicine [4–6].

Ref. [4] has demonstrated for the first time a high repetition-rate, table-top, laser-driven Proton-Boron Alpha-Source. Using laser pulse repetition rates of 10 Hz and 1000 Hz, Alpha-fluxes of 10^5 and 10^6 alphas/sec were measured, respectively. This is a record Alpha flux obtained from such ‘table-top’ high repetition laser-plasmas. The target was a BN plate coated with a thin layer of poly (ethylene). The target was moved between the 1000 Hz laser pulses, such that each laser pulse irradiated a fresh area of the target. Note that each laser focal spot on target is only few microns diameter. The laser driver has a wavelength of 1030 nm, maximum pulse energy of 20 mJ/pulse a pulse duration of 1.5 ps pulse and was focused on target at an intensity of $2.3 \times 10^6 \text{ Wcm}^{-2}$. This flux requires scaling by a factor of $10^5 \times$.

We could scale the Alpha-particle flux for example by increasing the: (i) Conversion efficiency of the laser energy into Alpha-particle-flux. We propose to achieve this by using the new, Borane target materials; (ii) Average laser power on target = (pulse energy) $\times$ (pulse repetition rate). At high repetition rates we also need to minimize target debris by using picosecond or femtosecond laser pulses, thin targets and low pressure buffer gas, as discussed in section 5.1 below.

Ref. [4] has shown how high repetition rate lasers enhance the Alpha-particle-flux even when ‘table-top’ laser drivers are used. But this requires high-repetition rate targets. We propose the following two such high repetition rate target concepts. These target concepts are widely used but we propose to make the targets from the new, Borane materials:

- (a) Liquid, molten material, droplet- [16, 17], or jet-targets. In [17] a >20 kW, 50 kHz industrial laser irradiates tin droplets at 50 kHz which emit EUV radiation in the industrial stepper for nano-meter EUV lithography. For Alpha-particle radiation sources, the liquids we propose are molten Borane material or Borane solution — many Boranes are soluble. Other possible such fluid targets could be: Borane gas jets, Borane nano-clusters. One could even consider Borane powder micro-crystals dispensed by a nozzle.
- (b) Tape targets [18, 19]. In [19] a 300 W, 300 Hz laser focused on a moving, 25 mm thick, Copper, tape-target generated 30 W of 1 keV Soft X-rays with a conversion efficiency of 10%. In order to generate Alpha particles we propose tape-targets covered with a Borane layer.

For high-repetition laser-plasma interactions, it is advantageous to introduce buffer gas in the interaction chamber (or the vicinity of the laser-target/plasma interaction volume) in order to stop the ‘ionic debris ejected from the target’ and therefore improve laser shot-to-shot energy deposition on target as well as to avoid coating the focusing lens with ‘atomic’ debris in long term operation. For liquid-Tin targets EUVL sources, very-low pressure Hydrogen-gas is used in industrial ‘steppers’ [17]. The low Hydrogen pressure is required in order to minimize the absorption of the EUV radiation by the buffer gas. Much higher buffer gas pressures can be used in laser-generated X-ray Sources from tape targets: in [18] one atmosphere, 1 Bar, of Helium buffer gas is used in the X-ray source generating ~ 1 nm X-ray photon energy from copper tape-target irradiated with UV, KrF laser driver (0.248 nm wavelength). In a similar laser driven copper-tape, ~ 1 nm X-ray Source described in [19], lower pressure, ~ 300 mBar of Helium buffer gas is used because the YAG laser-driver has a longer wavelength of 1024 nm compared to the KrF UV laser at 248 nm. Very short, picosecond laser pulses [18] also reduce the plasma debris.

Solutions to contain target debris in the case of laser-driven Alpha-particle sources would be somewhat different from those used in X-ray and EUV sources. The buffer gas will need be at very low pressure, mBar or sub-mBar. Otherwise the buffer gas would efficiently stop the Alpha-particles themselves. So very low pressure, sub-mBar Hydrogen gas could be tried, together with short pulse duration, picosecond or femtosecond lasers and thin, ~ 20 μm , targets. The presence buffer gas could deflect the P-B fusion Alpha particles from a ballistic trajectory and even stop them, together with the plasma ions. This could be avoided by using new geometries of Alpha particle interaction with materials we plan to irradiate, like radioisotope target material. We could mix the radioisotope target material with the Borane powder producing ‘mixed target’ as described in section 7.1. Or we could coat the material in a thin layer around the Borane P-B fusion target. This way the Alpha particles generated by in-target P-B fusion will produce nuclear reactions in the in-target radioisotope material, in the proximity of the Alpha emission. This geometry would avoid Alpha particles traveling long distances in buffer gas, even at very low pressure.

In order to quantify the Alpha-particle flux enhancement concepts we propose as an example a combination of parameters which could scale up the flux of Alpha-particles. Increase the laser repetition rate of to 50 kHz, 20 mJ/pulse, 2 kW, sub-picosecond/picosecond fibre laser, e.g. AFS Ytterbium-2000 laser [15]. Let us assume we could also achieve, with such a laser, the state of the art normalized Alpha flux of the order of $\sim 10^8$ Alphas/J/sr (figure 1). Under these conditions the Alpha source would generate and Alpha Flux (BN-target) ~ 50 kHz $\times$ 20 mJ/pulse $\times$ 10^8 Alphas /sr/J $\times$ 4 pi sr $\sim 10^{12}$ Alphas/second. We expect this flux to increase by more than 10-fold (section 2) to

Alpha Flux ($\text{BNH}_6\text{-target}$) $> 10^{13}$ Alphas/second by using, Hydrogen-rich Ammonia Borane molten droplet, tape or disc targets as proposed in the next subsections 5.1 and 5.2.

5.1 Proposed, new, molten Borane liquid-droplet or -jet targets for high repetition rate/high flux laser-driven Alpha-particle sources

Liquid droplets or jets are a convenient way of providing fresh targets for high and very high repetition rate lasers, figure 4(a). For example, piezoelectric-nozzle liquid droplet dispensers can deliver molten-tin droplets on which the focused laser generates a plasma emitting EUV radiation (13.4 nm wavelength) for EUV nano-lithography of semiconductor chips [16, 17]. Such industrial EUV Lithography Steppers with a Tin-droplet-laser-plasma-source operate at a repetition rate of 50 kHz [17].

We propose to use this technology to dispense molten Borane droplets or jets, in particular molten Ammonia Borane droplet or jet targets. The Boranes melt at a low temperature of $\sim 100^\circ\text{C}$ — Ammonia Borane melts at 104°C — which makes them ideal materials for liquid, molten targets. For comparison Tin used for EUV Lithography droplet targets, melts at $\sim 232^\circ\text{C}$.

Figure 4 is a schematic of the proposed liquid Borane targets, for example, molten Ammonia Borane material. The Borane target material is heated to $\sim 100^\circ\text{C}$ in the top vessel which supplies the molten-liquid to the nozzle. The droplets are dispensed by the piezoelectric vibrating nozzle with droplet repetition rates from tens of Herz to hundreds of kilo-Hertz. The droplet size can be selected by adjusting the nozzle-size, repetition rate and liquid pressure at the nozzle. For completeness, we will also consider using solid ‘cluster’ Borane targets dispensed by the nozzle.

The size of the molten-droplet can be adjusted from a few micrometres to hundreds of micrometres. For ‘low energy’ high repetition lasers, the smaller droplet diameters would be preferred because they generate less target debris for the same Alpha emission.

The droplets are irradiated from lateral laser(s) pulses focused on the droplets and synchronized to the droplet repetition rate or fractions of it, if the laser is lower repetition rate.

The laser-driver will be focused at the ‘optimum’ intensity for maximum Alpha particle generation. Ref. [4] uses a laser intensity of $2.3 \times 10^{16} \text{ W/cm}^2$ although high Alpha fluxes were obtained with both much lower and much higher intensities depending on the laser pulse used, as shown in introduction and figure 1.

The laser-plasma generated on the Ammonia Borane material induces the ‘in-target’ Proton-Boron fusion reactions which emit Alpha particles. The Alpha particle source-size is a small, ‘point-source’ positioned at the focus of the laser where the laser-droplet interaction occurs. Therefore the ‘brilliance’ of the Alpha-source will be enhanced by the small size of the source. The pulse repetition rate of the Alpha-particle source is the same as the laser repetition rate which is equal to the droplets repetition rate. For example 50 kHz as in [17]. The Alpha particle emission can be considered in the first order as distributed over 4-pi steradian centred on the laser focus: future experiments will establish the precise angular distribution of Alpha emission in such a geometry.

The waste liquid target material can be recycled by using heated collecting vessels to keep the waste target material in molten/liquid state (bottom). The waste liquid is pumped back to the liquid target reservoir.

In order to reduce the target debris from the laser-driven Alpha-source, we propose to use short pulse lasers, ps or fs, thin targets in low pressure buffer gas — possibly Hydrogen as in ref. [17].

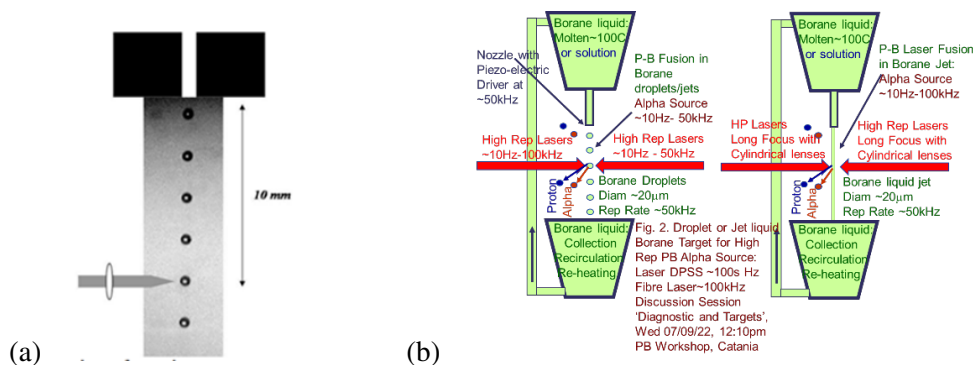


Figure 4. Molten material, liquid- and jet-Targets for high repetition laser-plasma radiation sources targets. The liquid targets are dispensed through a piezo-electric driven nozzle: (a) Molten Tin-droplet laser-target for EUV Lithography Source emitting 13.4 nm radiation. The laser is focused from the left on the Tin-droplet-targets, generating a Tin laser-plasma-source of EUV radiation. Reproduced with permission from ref. [16], C.-S. Koay et al., *High-conversion-efficiency tin material laser-plasma source for EUVL*, *Proc. SPIE* **5037** (2003) 801, in which one of us, ICET is co-author; (b) proposed molten Ammonia-Borane liquid-droplets or liquid-jets for high repetition-rate laser driven high-average flux Alpha Sources. Ammonia Borane has a low melting temperature of 104C. The waste liquid target material can be recycled by using heated collecting vessels (bottom) with filtered recirculation pump back to the liquid target reservoir. Buffer gas is present around the laser-target in order to contain the expanding target debris and hence avoid laser absorption of next laser pulse as well as debris contamination optics, detectors and Alpha-radiation exposure cells.

5.2 Proposed, new, Borane coated tape targets for high repetition rate/ high flux laser-driven Alpha-particle Sources

We propose several types of high repetition rate, laser-driven tape-and plate-targets containing Borane active materials for generation of repetitive Alpha-sources.

First, we propose to coat with Ammonia Borane powder the BN substrate plate used by [4]. We expect a more than 10-fold increase the efficiency of converting the laser energy into Alpha particles (section 2). We propose to manufacture the superficial Ammonia Borane layer, for example, by melting and solidifying the Ammonia Borane powder on the surface of the BN plate. Or simply gluing Borane powder on the substrate. There would be a problem with target debris if the plate is thicker than a few tens of micrometres. Ref. [18] shows that ‘thick targets’ or ‘rods’ emit large amount of ‘cluster-like’ debris which propagates on ballistic trajectories away from the target. This type of debris is greatly reduced when the solid target thickness is, say 25 μm or less [18].

Secondly, we propose the concept of a tape-target similar to that routinely used for X-ray generation from laser plasmas [18, 19]. For a repetitive Alpha-source we propose to cover for example a plastic tape-target with a layer of Ammonia Borane crystalline powder, possibly glued on the plastic tape, in a similar way to the Ferro-magnetic material in the old ‘tape-recorder’ tapes. Another proposed method would be to: (i) cover the plastic tape with a thin borane powder containing micro/nanocrystals; and (ii) seal the Borane powder on the tape substrate by coating a very thin plastic layer on top of the Borane powder. The laser will be focused on the Borane powder sealed on tape by the ultra-thin plastic. The Hydrogen and Boron ions will be accelerated inside the Borane plasma. They will collide and generate Proton-Boron fusion reactions producing Alpha-particles. The ‘tape-target’ will be translated at the speed required to provide a fresh target for the next repetitive laser pulse. This will result in a repetitive alpha source.

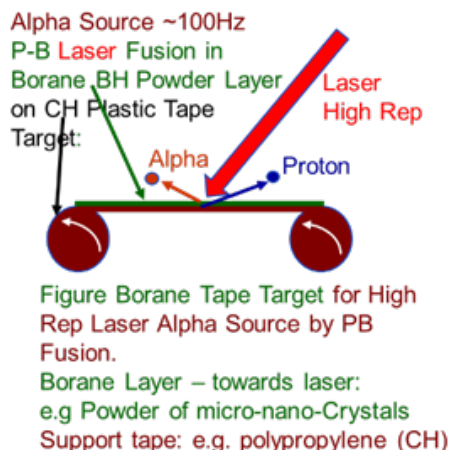


Figure 5. Proposed tape-target for Alpha generation from laser-driven Proton-Boron Fusion reactions in the Borane layer. The plastic tape target is covered with borane micro-crystals powder. The cartoon was presented by the author ICET during the “2nd International Workshop on Proton-Boron Fusion”, Catania, Italy, 2022, ref. [14]. Tape targets are often used for repetitive laser-plasma radiation sources. More details and pictures of ‘laser tape targets’ made of copper-foil and used for high average power soft X-ray generation can found in ref. [18], ICE Turcu and JB Dance, “X-rays from Laser Plasmas”, J Wiley, 1999, figure 7.3.

Figure 5 shows the proposed tape-target for Alpha generation from laser-driven Proton-Boron Fusion reactions in the Borane layer. The plastic tape target is covered with borane micro-crystals powder. The cartoon was presented by the author ICET during the “2nd International Workshop on Proton-Boron Fusion”, Catania, Italy, 2022, ref. [14]. More details and pictures of ‘laser tape targets’ made of copper foil and used for high average power soft X-ray generation can found in Ref. [18], ICE Turcu and JB Dance, “X-rays from Laser Plasmas”, J Wiley, 1999, figure 7.3.

6 Proposed new micro- and nano-structure Borane targets for studies of P-B fusion Alpha-sources driven by Petawatt lasers focused to very high intensities

The mechanism of laser plasma ion acceleration which results in such energetic Proton-Boron fusion reactions is not well understood: we obtain high yield of Alpha particles with lower laser intensity of $3 \times 10^{16} \text{ W/cm}^2$ from PALS laser but also at high intensity of $3 \times 10^{19} \text{ W/cm}^2$ from LFEX laser (section 1 and figure 1).

We propose to further investigate the effect of laser intensity using the extremely high intensity PetaWatt (PW) lasers available, which provide intensities of $\sim 10^{21} \text{ W/cm}^2$ and even higher in the near future. We could study the ion acceleration and P-B fusion reaction in ‘in-target’ experiments as a function of laser intensity up to such extreme laser intensities on target.

As the laser intensities increase above $\sim 10^{19} \text{ W/cm}^2$, the laser absorption mechanism in target changes and there is a reduction in the laser absorption in flat solid targets. A solution is to use Micro- [21] and Nano-structured [22–24] targets which show absorptions close to 100% even at such high intensities.

The nano- of micro-structured target would have the additional advantage that the laser Electro-Magnetic field could penetrate deeper in the target. This could lead to further enhance the state of

the art yield of P-B fusion reactions beyond the >10 -fold we expect from BNH6 target materials with smooth surfaces, as discussed in section 2.

Therefore in order to study P-B reactions at very high laser intensities we propose to use new geometries of Micro- and Nano-structured solid targets made of the new Borane materials, for example Ammonia Borane (section 2). We propose to use ‘Nano-structure’ Borane targets for PW lasers with Femtosecond pulse durations and ‘Micro-structure’ Borane targets for PW laser pulses with Picosecond duration. This is because the Nano-structures would be destroyed by pulses of longer duration.

6.1 Borane nano-wire targets for P-B Alpha-sources driven by femtosecond PetaWatt laser pulses

Laser irradiated Nanowire arrays generate extremely dense and hot plasmas when irradiated in the ‘Nano-Z-pinch regime’ as shown in the simulations of [24]. The nano-wire array consists of carbon nanowires of 300 nm diameter and 5 μm length. They are irradiated with a circularly polarized, 60 fs pulse duration, 400 nm wavelength laser focused on the nanowire array at relativistic intensity of $4.95 \times 10^{21} \text{ W cm}^{-2}$. The very high currents and magnetic fields induced in and around the nanowire compress and heat the plasma at very high densities and temperatures. Electron densities reach $9.4 \times 10^{24} \text{ cm}^{-3}$. For the highest laser intensities the electron temperature reaches 4 MeV and the carbon ions gain energies $> 100 \text{ MeV}$.

This special combination of very high pressure and temperature can be used to generate a nuclear fusion reaction. Indeed, [22] reports Deuterium-Deuterium (DD) fusion reactions in deuterated-polyethylene (CD_2) nanowire arrays irradiated in the ‘Nano-Z-pinch regime’. They measure a fusion neutron production of 2×10^6 neutrons/(Joule of laser pulse). The 400 nm, 60 fs, laser pulse energy of 1.64 J/pulse was focused on the nanowire array.

At laser intensities even higher than the ‘Nano-Z-pinch regime’, the laser electro-magnetic field can actually penetrate inside the nanowire array and indeed into the substrate material because the laser-plasma interaction occurs in the ‘relativistic transparency regime’ as shown in [23] and [25]. Ref. [23] has irradiated deuterated-polyethylene (CD_2) nanowire arrays at laser intensities of $3 \times 10^{21} \text{ W cm}^{-2}$ and measured 1.2×10^7 neutrons/pulse generated by DD fusion reactions. The laser pulse energy was 8 J/pulse. They also measured energetic ion-beams accelerated to energies of 13 MeV in the nanowires in the directions opposite to the laser pulse as well as 20 MeV protons accelerated in the direction of the laser and into the substrate material.

We propose to generate such Proton-Boron fusion reactions in Borane nanowire array plasmas in both the ‘Nano-Z-pinch regime’ and the ‘relativistic transparency regime’, depending on the laser-intensity on target and target density. Both laser-plasma regimes generate and exceed the Proton-Boron fusion requirements. The ions have kinetic energies around 1 MeV. The plasma densities are high, close to solid densities. We will replace the deuterated-polyethylene (CD_2) material in the nanowires with Borane material, like Ammonia Borane (BNH6), or even polyethylene (CH_2). We also propose to replace the substrate material with solid Boron, Borane or Boron Nitride.

Figure 6 and figure 7(1) show proposed new target geometries for Borane nano-structured targets for PW, Femtosecond laser driven PB fusion alpha-sources. In figure 6. the Nano-Wires [NW] are made of Borane compounds, e.g. Ammonia Borane (BNH6) or plastic, e.g. polyethylene (CH_2). The nanowires are supported on a Boron, Borane or Boron-nitride substrate. Nanowires are irradiated by PW, Femtosecond Laser. The energetic and dense ions accelerated by the laser EM

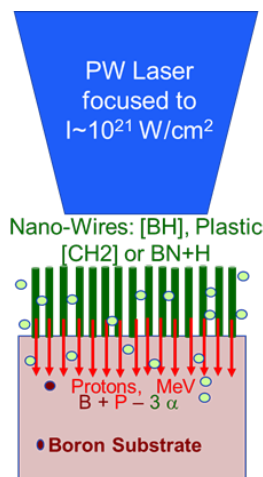


Figure 6. Proposed, new Nanostructures Proton-Boron-fusion Targets: Nano-Wire are made of Boranes (BnHm) and compounds, e.g. Ammonia Borane (BNH₆), or plastic, e.g. polyethylene (CH₂). The nanowires are supported on a Boron, Borane or Boron-nitride substrate. Nanowires are irradiated by PW, Femtosecond Laser. The energetic and dense ions accelerated by the laser EM fields collide between themselves and with the substrate nuclei therefore emitting Alpha-particles from Proton-Boron fusion reactions.

fields collide between themselves and with the substrate nuclei therefore emitting Alpha-particles from Proton-Boron fusion reactions.

The nanowires in figure 6 could also be made of plastic (e.g. CH₂). This is because a large number of ions and protons are accelerated towards the substrate in the experiments reported in both the ‘Nano-Z-pinch regime’ Curtis (2018) and the ‘relativistic transparency regime’ Curtis (2021). The papers comment that the number of nuclear reactions in the substrate could exceed the reactions in the nanowires themselves. Therefore it would be interesting to study Alpha-Emission from targets with a combination of plastic (C₂H₂) which, under laser irradiation, inject great number of energetic protons in the Boron substrate.

We also propose to irradiate a powder of randomly arranged nano-crystals of Borane (e.g. Ammonia-Borane). There is the assumption that the random nano-crystals will also undergo large laser absorption and proton acceleration by EM forces similar to those occurring in ordered nano-wire arrays, oriented towards the laser, as in [22–24]. The nano-crystal powder is contained in a hole in the Boron or Borane substrate as shown in figure 7(1). Several vendors offer such ‘nanocrystal’ Borane powders as compared to just mixed nano- and micro-crystal Borane powders.

The focussed drive laser will irradiate the nano-crystal powder and will be efficiently absorbed. The ions and protons will be accelerated in the nano-crystals by the laser field, and will collide between themselves in the nano-crystals. A large number of accelerated ions and protons will emerge from the irradiated crystals and collide with the Boron nuclei in the walls of the hole containing the powder (figure 7(1)).

The diameter of the hole containing the Borane powder hole can be several microns or tens of micron diameter, depending on the size of the laser focal spot. The hole in the Boron substrate could have several shapes/geometries in order to maximize the number of nuclear reactions, for example: (a) cylindrical filled with borane crystal powder; (b) cylindrical with a thin Boron cone (or cylinder) inside the containing hole — the Borane crystal powder will fill the volume between

the central Boron cone/cylinder and the Boron walls of the containing hole in the substrate; (c) any geometry which maximizes the Boron/Borane covered area in the direction of all the accelerated ions and protons emerging from the Borane crystal powder.

6.2 Borane micro-structure targets for P-B fusion Alpha sources driven by picosecond PetaWatt laser pulses

We propose to use microstructured-Borane-targets for Petawatt laser pulses of higher energy/pulse (e.g. kJ/pulse) but longer pulse durations, of picoseconds/pulse instead of femtoseconds/pulse. At such longer pulse durations the nano-structures would destruct during the laser pulse. This makes the micro-structures optimum targets while still showing high absorption for very high intensity lasers pulses.

For example, [21] measures $\sim 95\%$ laser absorption when focused at high intensities, $I \sim 2 \times 10^{20} \text{ W/cm}^2$ on ‘micro-cone’ silicon targets. The Silicon ‘micro-cone’ targets consisted of an array of cones $15 \mu\text{m}$ high and with $5 \mu\text{m}$ diameter supported on a $5 \mu\text{m}$ thick substrate. The laser wavelength is $1.053 \mu\text{m}$, energy per pulse is $\sim 160 \text{ J/pulse}$, pulse duration $\sim 1 \text{ ps}$ and focal spot diameter $\sim 10 \mu\text{m}$. When the shorter $0.527 \mu\text{m}$ laser wavelength is used, the laser absorption in the target reaches $\sim 100\%$.

High energy electrons, reaching $>30 \text{ MeV/electron}$ with a 4.6 MeV ‘electron temperature’ are measured especially in the trough between two adjoining micro-cones. Large energy protons are accelerated from the back of the substrate, with a proton ‘temperature’ of 2.36 MeV . Note this is much higher than the 0.7 MeV kinetic energy required to achieve maximum reaction crosssection in Proton-Boron fusion. The measured number of protons accelerated in the micro-cone targets is 1.9×10^{12} protons/pulse. A high energy conversion efficiency of 1% from laser pulse energy to proton pulse energy is measured from protons accelerated to energies above 4 MeV/proton . The conversion efficiency would be higher if not limited by the number of hydrogen atoms in the contamination layer at the back of the substrate.

In view of such remarkable laser absorption and proton acceleration in micro-structured targets, we propose to study Proton-Boron reactions in such micro-structured targets but made of Borane material instead of Silicon.

Figure 7 shows our proposed new target geometries for Borane micro-structured targets for PW laser driven PB fusion alpha-sources. Most solid Borane materials would be suitable but we prefer to use the particular Borane: Ammonia Borane, for the reasons shown in sections 2 and 3.

In the design of figure 7(1) we propose a new micro-structured target made of Borane micro-and-nano-crystals powder contained in a hole in the Boron substrate. The hole can be several microns or tens of micron diameter, depending on the size of the laser focal spot. The laser-driver is focused on the crystal powder in the hole and is absorbed efficiently in this random micro/nano-structure. Electrons and ions will be accelerated in the crystals forming the powder. Since Borane is rich in Hydrogen and Boron, the accelerated Protons and Boron ions will collide in fusion reactions which generate Alpha particles. Many accelerated ions will also escape the laser-plasma formed by the powder target. These protons will collide with the Boron ions in the laser-plasma formed on the Boron walls of the containing hole resulting in more PB fusion reactions and Alpha-particle generation. Such a target would be easy to manufacture both (i) as a ‘single-shot’ laser target; and (ii) as repetitive laser target made of a long Boron or Boron-nitride plate with many holes filled with Borane powder. The plate is translated to a fresh hole between two laser pulses.

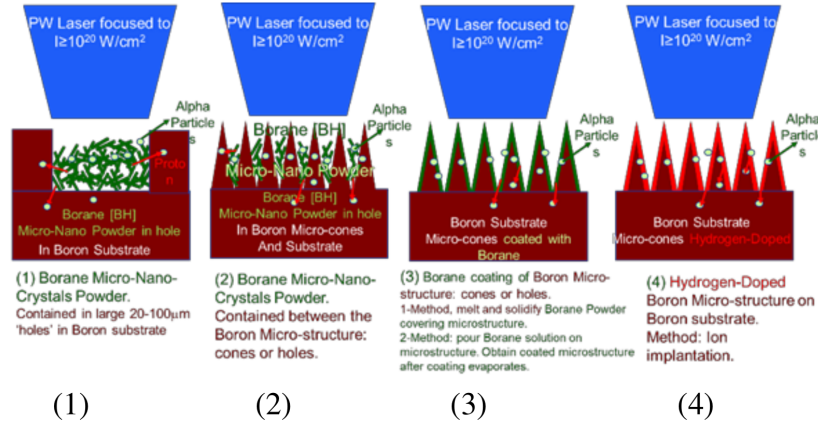


Figure 7. Proposed, new Borane Micro-structured targets for PW Laser driven PB fusion Alpha-sources: PB fusion in Borane crystals and in Boron substrate from energetic protons. (1) Borane micro-nano-crystals powder contained in a hole in the Boron substrate. The hole can be several microns or tens of micron diameter, depending on the size of the laser focal spot. (2) Boron micro-cone-array with the gap between the cones filled with Borane micro-nano-crystalline powder. (3) Boron Micro-cones coated with Borane layer. The Boron micro-cone-array could also be conical holes drilled in the boron substrate, again coated with Borane. (4) Boron micro-cone arrays enriched by Hydrogen ion implantation. Also conical holes as in (3) implanted with hydrogen. Most solid Borane materials would be suitable but we prefer to use the particular Borane: Ammonia Borane, for the reasons shown in sections 2 and 3.

In the design of figure 7(2) we propose a new target similar to 6.1 but instead of a cylindrical containing hole, the Borane micro/nano-crystal powder is contained in the gaps between micro-cones made of Borane, Boron or Boron-nitride. Such a target would operate similarly to target 7.1 under laser irradiation but would maximize the ion collisions between the ions accelerated in the borane crystals and the Boron ions from the plasma formed on the containing walls.

In figure 7(3) and 7(4) we propose two new designs of micro-structured targets made of Boron Micro-cones either coated with Borane layer or solid Boron with a surface laser enriched in Hydrogen by ion-implantation. The Boron micro-cone-array could also be conical holes drilled in the boron substrate, again coated with Borane or implanted with hydrogen atoms. The substrate would be made of solid Boron. The Borane coating of the Boron micro-cones may be achieved by (i) covering the cones with a Borane powder; (ii) melting the powder at low, $\sim 100^\circ\text{C}$ temperature and (iii) cooling the micro-cone array and the molten Borane which may result in a solid Borane glazing of the micro-cones. When irradiated by laser the Borane coated micro-cones will absorb efficiently the high intensity laser. Ions will be accelerated in the target, will collide between themselves and with substrate ions and atoms. The collisions leading to Proton-Boron fusion reactions will generate a large amount of Alpha-particles.

7 Proposed applications of Borane targets to medical radioisotopes and nuclear-fuels for fusion energy generation

7.1 Proposed medical radioisotopes fabrication schemes using laser-driven nuclear reactions within Borane targets

Several medical radioisotopes/radionuclides can be manufactured by irradiation with Alpha particles of a target material [5]. Such radio-isotopes have many applications in both: medical diagnostics,

for example Positron Emission Tomography (PET); or in targeted, in-body radiotherapy for example of cancer tumours.

The laser-driven P-B fusion Alpha source could for example be deployed in Hospitals for manufacture of short lived radio-isotopes which need be manufactured in-situ to avoid the transportation time needed if manufactured at a central accelerator facility.

For example let us consider the production of the positron emitter radionuclides ^{11}C ($T_{1/2} = 20.4$ min) or ^{18}F ($T_{1/2} = 109.8$ min), which are commonly used in Positron Emission Tomography (PET), respectively [5]. The laser irradiates say the Ammonia Borane, BNH_6 , solid-pellet-target (section 4) or, high repetition rate laser irradiates molten-droplets/jets (section 5.1) or powder-coated-tape target (section 5.2). The in-target accelerated Protons collide with the Nitrogen nuclei to form the ^{11}C radionuclide: $^{14}\text{N}(\text{p}, \alpha)^{11}\text{C}$ [6]. The in-target, P-B fusion generated Alpha particles could also collide with the Nitrogen nuclei to form the ^{18}F radionuclide through the reaction $^{14}\text{N}(\alpha, \gamma)^{18}\text{F}$.

The radionuclide ^{30}P ($T_{1/2} = 2.5$ min) is an interesting short-lived non-standard positron emitter with application in biological studies both in animals and humans using PET [5]. It can be produced via the nuclear reaction $^{27}\text{Al}(\alpha, \text{n})^{30}\text{P}$ which can be induced by the low-energy (5.3 MeV) α -particles. Note that this low energy Alpha particles are naturally generated as a result of P-B fusion. We could propose irradiating with laser a mixed target of compressed Ammonia Borane and Aluminium filings. The in-target Proton-Born fusion reactions generate multi-MeV Alpha particles which can collide with the Al nuclei to generate the ^{30}P radionuclide.

7.2 Proposed Borane-nuclear-fuels for laser-driven Proton Boron laser-fusion energy production

There is great interest in generating energy from nuclear fusion reactions of Proton-Boron nuclei by heating the fuel to high temperatures [7–13].

The reaction is aneutronic [26] and therefore the reactor generates very low radioactivity. The fusion energy is released as kinetic energy of the three charged particles (Helium nuclei). The energy of charged particles can be efficiently transformed directly into electricity by Electric and Magnetic fields.

On the other hand P-B fusion energy will be harder to obtain because: (i) The fusion cross-section of the Proton-Boron reaction is maximum for very high ion kinetic energies of ~ 600 keV compared to the Deuterium-Tritium fusion cross-section which require much lower, ion kinetic energies of ~ 15 keV; (ii) The energetic electron in passing a Boron ion in the plasma, loses $25\times$ more radiation energy (Bremsstrahlung) than when it passes a Hydrogen ion. (iii) The fusion energy generated by P-B fusion reaction is 8.7 MeV/reaction which is $\sim$ half that from D-T which is ~ 17.6 MeV/reaction.

Let us briefly discuss potential advantages which could make Boranes interesting new fuel materials for laser-driven P-B fusion energy reactors.

Boranes contain Boron and Hydrogen the two elements which can produce Proton-Born Fusion.

Boron compounds are readily available on Earth and have wide ranging applications in society and industry. Boranes can be obtained from other compounds by simple chemistry. Some Boranes are solid crystalline powder at room temperature and pressure, compared with heavy Hydrogen fuel which requires cryogenic cooling. For high repetition rate laser-fusion the targets could be dispensed as molten droplets or solid pellets.

For laser-driven thermal-plasma P-B fusion energy production, ref. [13] requires a fusion fuel with a ratio of B/H ~ 0.15 or H/B $\sim 667\%$. Such a high ratio of Hydrogen to Boron density will contribute

to reduce both the effective ionic charge and the total number of electrons in the plasma (each fully ionized Boron ion contributes five electrons). This in turn reduces the Bremsstrahlung emission as well as the energy lost in heating the extra electrons. Also, a ratio B/H around 0.3 maximises the yield of suprathermal fusion induced by the generated alpha particles [12].

To satisfy the above requirements for P-B fusion fuel, the Borane molecule with the maximum Boron dilution in Hydrogen would be Diborane (6), B_2H_6 , which has ratio of $B/H = 0.33$. Additional methods to dilute the concentration of Boron in hydrogen will be necessary to achieve $B/H \sim 0.15$, as required above. In Diborane (6) the Boron dilution B/H ratio is higher than for Ammonia Borane (BNH_6) molecule ($B/H = 0.17$) which nevertheless also contains Nitrogen which contributes negatively to the above requirements of Bremsstrahlung and total electrons in plasma. Diborane (6) is a toxic and pyrophoric gas in normal conditions. It becomes liquid at temperatures below -92.49°C and becomes solid at temperatures below -164.85°C . Another potential P-B fusion fuel Borane candidate would be Tetraborane (10), B_4H_{10} . Tetraborane (10) has a ratio $B/H = 0.4$, and therefore would also require additional methods to dilute the concentration of Boron in hydrogen. Tetraborane (10), although still toxic, has higher boiling point of 18°C and higher melting point of -120.8°C compared to Diborane (6).

8 Conclusions

We propose, for the first time, a new fusion-fuel material, Ammonia Borane (BNH_6), to enhance by more than tenfold the yield of Alpha-particles generated by laser-driven Proton-Boron (P-B) nuclear fusion reactions compared to state of the art Boron Nitride (BN) target material. The Ammonia Borane has one thousand-fold larger Hydrogen/Boron concentration of $H/B = 600\%$ compared to Boron Nitride with $H/B < 1\%$.

We have manufactured the first Ammonia Borane pellet laser-targets, by mechanically compressing commercially available powder containing micro- and nano- crystals.

To further increase the Alpha particle flux, we propose new target geometries for laser-drivers with increased repetition rate, to kHz and above. We propose, Ammonia Borane targets in the shape of liquid jets or high repetition droplets, as well as fast translating coated tape-targets.

In order to study and optimize ultra-intense, PetaWatt laser-driven P-B fusion reactions we have proposed nano- and micro-structured Ammonia Borane target geometries. Such targets are expected to enhance the yield of nuclear fusion reactions by providing improved laser absorption, deeper in the target as well as high particle acceleration to large kinetic energies.

We propose several applications of the laser-driven Borane targets to medical radioisotopes and nuclear-fuels for fusion energy generation.

We propose medical radioisotopes fabrication schemes using laser-driven nuclear reactions within Borane targets mixed with radioisotope target material. This new target geometry could potentially enhance the yield of the radioisotope material due to the proximity of the energetic alpha or proton particles to the radioisotope target material. To improve the logistics of short-lived radionuclide delivery to the patient, laser-driven radioisotope production beamlines would be installed in hospitals.

We also propose Boranes, for example Diborane (6), B_2H_6 , as a nuclear-fuel for laser-driven Proton-Boron fusion energy generation schemes. The small Boron/Hydrogen ratio of $B/H = 33\%$ may need be further diluted with Hydrogen to achieve $B/H < 15\%$ in order to cut the Bremsstrahlung X-Ray radiation losses and other deleterious losses which could prevent ignition of the hot, dense fusion pellet.

Acknowledgments

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